

Adaptive and Argumentation-based Human-Agent Dialogues

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Abstract —

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1 Introduction

Assistive technology aims to support an individual in accomplishing activities, which they need to be able to do in the presence of decreased functionality or ability. In this work the definition of *active* assistive technology used by Kennedy and co-workers [1] is applied to distinguish systems, which includes automated processing of health information during a human agent's interaction with the system and which may output tailored responses to the human agent in the process. The purposes of active assistive technology include the following: to increase knowledge, assist in deciding about actions to make, and promote changes of unhealthy behavior (i.e., in the form of *behavior change systems*).

The aim is to enable dialogues between a human agent and a software agent, which are tailored to the human's needs, preferences and goals. For this purpose, the software agent needs the following capabilities, which has to different extent also been described in literature (e.g., [2–6]):

1. Being autonomous (e.g., decides upon actions to make, takes initiatives in order to reach a goal),
2. Utilize a *semantic model* shared between the human and the agent for communication and knowledge exchange purposes,
3. Generate new knowledge and learn,
4. Being cooperative,
5. Being able to deal with affective components and topics in a dialogue,
6. Being adaptive to the human's needs, preferences and the situation.

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Typically, a team of healthcare professionals can be involved in an elderly person's care, such as doctors, nurses and occupational therapists. These together with the human cooperate to gain optimal understanding of the human's health condition in a process of *assessment*. Consequently, there is a need for combining information from different sources to understand the external factors affecting the human. The support provided to the human is aimed to be person-centered, and the assessment includes deciding upon what *interventions* may be implemented for improving the human's situation.

A prototype dialogue system was implemented, which enables the human actor to conduct dialogues on different health-related topics with the Coach Agent in their home environment. The prototype was evaluated in a pilot study involving a group of therapists. The purpose was to evaluate from healthcare experts' perspectives, the agent's behavior when conducting the dialogue, its use of the domain knowledge and user model.

This paper is organized as follows. In the following section the methods are described and in Section 3 an analysis of related work is provided. In Section 4 a model of the Coach Agent is provided. Section 5 describes the pilot evaluation study. The results are discussed and the article ends with conclusions and comments on future work.

2 Methods

A literature study was conducted where the following topics were studied: i) the agent's role, ii) the purpose of dialogue systems, iii) the knowledge model of the agent, the user, the domain and the dialogue activity, iv) who develops the knowledge model, v) sources for the knowledge, vi) representation format and how generic the representation is, and vii) issues regarding the dialogue execution.

A pilot evaluation study was conducted involving a group of female professionals in occupational therapy and physiotherapy, specialized in the needs of older adults. The study was formative, with the results aimed to inform further development.

The main research question addressed in this paper is that, can the use of personalised argumentation dialogues with older adults on health-related topics supplement to the healthcare professionals future intervention?

To answer this research question, we design and develop a goal-directed semantic dialogue activity model. This model was evaluated by professionals in occupational therapy and physiotherapy. The sub-research questions, to answer the main research question, were the following:

1. Was the outcome of each argumentation dialogue satisfying the requirements expected by the professionals?
2. What new information can be added to the knowledge base?
3. Which features of the user interaction can be added or enhanced?

During the evaluation study, the healthcare professional tested the dialogue system by initiating a dialogue with the Coach agent. As the dialogue unfolded, the professional's dialogue moves on the screen and their running comments were observed and noted down. When the evaluator observed the end of each type of dialogue, asked the following questions:

1. End of Information-seeking dialogue
 - (a) Were the questions asked by the agent considered as appropriate to the topic?
 - (b) Was there any other question the professional would like the agent to ask?
 - (c) Was the user-interaction design for answering the questions intuitive?
2. End of Inquiry dialogue
 - (a) Did the professional and agent find a new knowledge about the older adult's health condition?
3. End of Deliberation dialogue
 - (a) Was there a plan of action agreed upon between the participants of the dialogue?
 - (b) Any new actions that can be added to the knowledge base?

Observation of use and open-ended interviews were the methods used for collecting data. The data was analysed qualitatively using content analysis.

3 Related Work

Research literature shows an increasing number of applications of software agents that interact with the human actors for health-related purposes (e.g., [7,8,5,9–11]). A review of behavior change systems utilizing personalization technologies for accomplishing active assistance is provided in [1]. It was concluded that an agent provides an effective interface modality especially for the applications that require repeated interactions over longer period of time, which is crucial for applications supporting behavior change [1,12].

One example is the system, which has been developed for older adults with cognitive impairment described in [7]. However, it focuses mainly on generating reminders about the activities of their daily living and takes no part in a complex dialogue with the user. Agent-based systems which interact with human actors through dialogues are less common. They are typically developed for a specific task or for a limited domain, and the dialogues are tested in specialized environments [11,9].

Bickmore et. al [11] developed an animated conversational agent that provides information about a patient's hospital discharge plan in an empathic fashion. The patients rated the agent very high on measures of satisfaction and ease of use, and also as being the most preferred over their doctors or nurses in the hospital when receiving their discharge information. This was related to the agent's capability to adapt to the user's pace of learning and giving information in a nonjudgmental manner. Another reason for the positive results was the amount of time that the agent spent with users helped them to establish a stronger emotional bond.

In the counseling dialogue system discussed in [12], the agent performs motivational interviewing with its knowledge about behavioral medicine and expresses empathy in dialogues. It provides the human agent with advice about exercise and healthy actions and follows up by asking the human followup questions. However it limits its services to counseling only and does not reason with evidence about the human's change of behavior based on information about their activities.

The agent-based approach to diagnostic decision-making presented in [13] has the potential to support the medical professionals in improving their diagnostic

assessments of patients. The purpose is to guide the user in the diagnostic reasoning and support learning and a change of behavior. However, the user's active participation in the dialogue is limited.

To summarize, the most common Agent roles, and consequently, the purposes of the exemplified dialogue systems are the following: i) promoting the human actor to maintain control, learn and change behavior, ii) to guide in reasoning, decision-making and other activities, and iii) to inform in an emphatic and patient way, and behave in a way tailored to the human agent's pace, needs and preferences.

The canonical model of an agent described by Fox and coworkers [3] is an extension of the Belief-Desire-Intention (BDI) model of an autonomous agent (e.g., [2]) and provides formal semantics for the concepts argumentation, decision-making, action, belief, goal (desire) planning (intention) and learning for agents. Literature describes implementation in systems, however, mainly for multi-agent reasoning and decision making without active participation of a human agent in the process [3]. To our knowledge, this model has not served as a base for dialogues between software agents and human agents beyond simple alerts or providing decisions.

The architecture of the agent-based ambient assistive living system presented in [14,15] integrates knowledge repositories, developed using ACKTUS (Activity-Centered modeling of Knowledge and interaction Tailored to USers), which is a platform for end-user development of knowledge-based systems in the medical and health domain. These repositories are built on semantic models represented using RDF/OWL, which were in [16] extended to integrate components vital for enabling the agent dialogues, based on the canonical model for agent architectures presented by Fox and coworkers [17]. This work led to a multi-agent system, which simulates a dialogue between an agent representing a physician who is the user of the system and who selects a diagnostic hypothesis in a patient case, and an agent representing a domain expert [13]. The domain knowledge and patient knowledge was used for creating arguments, based on sources with different strengths, which the agent-system evaluated for providing the user support in diagnostic decision-making. The domain knowledge was in this case retrieved from clinical practice guidelines, consensus guidelines, and other medical sources.

The ACKTUS core ontology was further extended to incorporate information about a user's environment, for the purpose of forming a user model, which an agent can use for adaptation of support in daily activities [18].

An initial design and implementation of an agent-based dialogue system where an agent communicates with the end user in a home environment on health-related topics is presented in [19]. The agent's belief base was created by obtaining information from the ACKTUS knowledge repositories, which had been created by domain experts in the rehabilitation domain [20]. The knowledge consisted of both factual knowledge related to a user, which was used for creating a baseline user model, and procedural knowledge, in the form of rules obtained from domain knowledge repositories. Moreover, interactive templates, or protocols for assessment of different aspects were retrieved from the domain knowledge repositories. The knowledge had been structured and implemented by the domain experts, to be used as assessment protocols by therapists in clinical interviews, or in human-agent dialogues.

The counseling dialogue system discussed in [12] also uses an ontology to represent the user model and the domain knowledge. The domain knowledge is based on behavioral medicine. Their approach consists of six models: theory model, be-

havior model, protocol model, user model, external data model and a task model. The authors have themselves developed the ontology based on a review of concepts in behavioral medicine, prior work, and discussions with experts in behavioral medicine, and computerized interventions. Their task model consists of the dialogue activity information with dialogue actions. However, the domain experts are not directly involved in the modeling of the agent's knowledge in this system therefore it lacks features for modifications such as future changes in domain knowledge and also it lacks the representation of goals which drives the behavior of the dialogue system.

Only a few applications developed for behavior change integrated into the system semantic models grounded in theories relevant for behavior change [1].

In argumentation literature there are several theoretical frameworks, which aim to formalize dialogues of different types for multi-agent systems. Typically, these adopt the approach of *dialogue games*, since they aim at selecting a "winner" of a dialogue, and then the game rules aim to create the dialogue game a fair competition (e.g., [21]). Based on argumentation schemes, dialogues can be built, e.g., for reasoning about actions to make related to the value of conducting a particular action [22]. Theoretical frameworks have been formalized also for handling nested dialogues of different types (e.g., [21]). However, these are typically not tested with dialogue information from real situations, and consequently, their applicability in situations such as our use case remains to be evaluated.

The base for interaction in [19] was implemented as information-seeking dialogues through which the user can inform the system about preferences, priorities and interests. Based on this information, the system provides feedback to the user in the form of: 1) suggestions of decisions, 2) advices and 3) suggestions of actions to make for obtaining more knowledge about a situation. However, to accomplish other types of dialogues, which are less structured than the *information-seeking* dialogues, an agent architecture is needed, which allows the agent to plan the moves based on e.g., the purpose of the dialogue, the situation, the knowledge available, etc. Therefore, this paper extends earlier work, by developing the components for constructing an intelligent software agent that interacts with a human in a more autonomous, adaptive and purposeful way.

Typically, a formal argumentation framework provides the semantics for solving conflicting arguments for or against a particular topic, regardless what the topic is. The effort to establish an *Argument Interchange Format* (AIF) for exchanging arguments on the World Wide Web is such example [23]. AIF is presented as an ontology, which takes different types of argumentation schemes, scheme-implementation nodes and their relations into consideration, without relating these to a concept-node system for topics. Most frameworks also do not consider vagueness, or uncertainty, or how these factors may change depending on the context of reasoning. Some theoretical work on argumentation frameworks consider *preferences* and *audiences* (e.g., [24–27]), which relate to the context of reasoning. In the presentation of AIF it is mentioned that the nodes may incorporate values, e.g., representing uncertainty, and domain knowledge, but not how such information is included in a formal way. In this work, there is a need to take such factors into consideration in dialogues. Therefore, AIF is applied, and extended with domain knowledge in the form of a concept-node system and templates for assessments of features with associated sets of allowed values. The approach to argumentation dialogues is still generic, since the knowledge domain can be replaced, depend-

ing on the topic of interest. This will be illustrated in the pilot study, where the agent retrieves topics from different domains of knowledge, represented as different domain knowledge repositories within the ACKTUS architecture.

Add Hunters articles [28,29]: e.g., our boolean “do you agree?” compared to Hunters probabilistic degrees of beliefs.

Finally, the need for *adaptation strategies*, which enable dynamic dialogues following the human agent's line of thinking is illuminated by the scenario.

The user and domain models are extracted from the ACKTUS ontologies [30, 18] in the dialogues between human and software agents. It should be noted that the domain knowledge repositories are structured in a way so that there are knowledge, which can be considered common, and other which is advanced expert knowledge, corresponding to a medical expert's knowledge. The Coach Agent integrates the part that is common knowledge into its belief base in dialogues in our implementation. The model of the Coach Agent have been developed as a part of this work, and presented in the following sections.

4 Coach Agent Model and Implementation

To conduct dialogues similar to natural dialogues, the Coach Agent was designed as an actor with its own knowledge base, goals and priorities, represented as an ontology, extended from the ACKTUS core ontology. There is for instance, behavioral knowledge, which is generic regardless which type of dialogue or topic is at focus. Emphatic behavior is one such type of generic social knowledge fetched from the agent's repository, since this knowledge needs its own semantic model, yet related to the human actor's, the Coach Agent was modeled as a separate project in ACKTUS (Figure 1). As a consequence, the Coach Agent shares the core ontology with the human and the domain knowledge models. This functions as the common vocabulary in dialogues.

The Agent model contains structures, which function as tools for the Coach Agent to use in the adaptation of its behavior. These tools aid the agent to perform collaborative human-agent activities which in our case is dialogues. (4.2.3) [ref ECCE instruments paper].

4.1 System implementation

The Human-Agent Dialogue(HAD) system has been implemented in Java as backend with HTML, JSP, jQuery and AJAX as frontend. The communication between a human and the Coach Agent happens as JSON messages. The HAD system is embedded into iHelp application. The iHelp is a prototype application for the older adults.

4.2 Heuristics for agent behaviour in human-agent dialogues

4.2.1 Rules for Dialogue Moves

A software agent must follow certain rules when having a dialogue with a human. In dialogue games, there is a restriction that agents take turn in the dialogues,

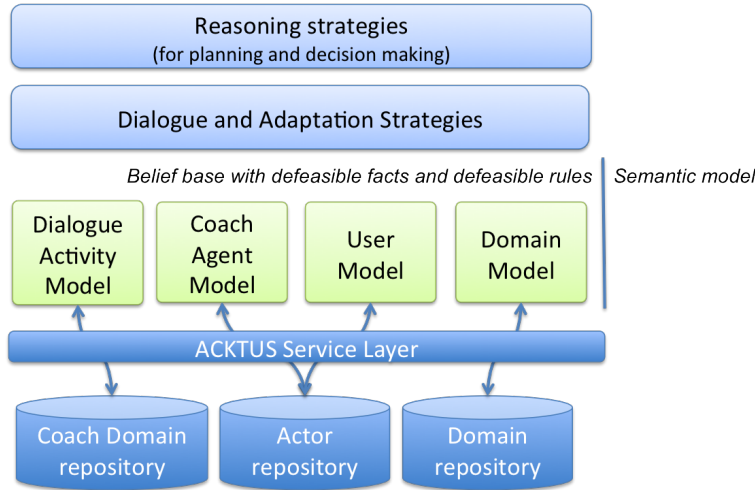


Fig. 1 The agent architecture, containing the Coach Agent’s belief base consisting of four models, and its different sources of knowledge.

and are allowed to perform only one move at the time. This rule does not apply in our dialogues. The reason is that the agents are not competing for “winning” the dialogue, and natural dialogues do not follow this restriction.

Other restrictions, such as the open move for a particular dialogue needs to take place before a close move, apply. Similarly, in case a pause move is done, then no move can be done within this dialogue until the dialogue is resumed by a resume move.

Only one dialogue can be active at a particular time point. This means that when a new dialogue is initiated by an open move, and if there is an ongoing dialogue, then this ongoing dialogue is paused. This follows the notion of “focus shifts”, which occur when humans conduct an activity, a phenomenon used for evaluating interactive systems.

Since it is the Coach Agent, which is directing the ongoing dialogue, the agent is also closing the dialogue. However, the human agent can close the main dialogue, and consequently all sub-dialogues, by switching the topic of the main dialogue.

Some moves have the purpose to defer the responsibility for acting to another agent. Such moves are *ask*, where the asking agent expects a response, and *alert*, when the alerting agent expects the other agent to take action since the situation requires action. Similarly, whenever the human agent responds to an ask move by a tell move, the responsibility to act is passed to the Coach Agent. Consequently, the agent needs strategies to handle the situation when the other agent does not respond as expected 4.2.2.

4.2.2 Adaptation Strategies

The protocol for conducting dialogues described in Section 4.2.1 contains a limited set of dialogue rules for handling basic moves in a dialogue. However, when conducting a dialogue with a human, the agent needs adaptation strategies to enable the human-agent collaboration to happen.

The agent uses the same strategy for selecting the next move in the different types of dialogues. The agent retrieves the set of information, which relates to the topic, and organizes these into a plan following the human's preferences among the topics. At each step in a dialogue when receiving new information from the human, the agent analyses the information, decides whether to continue the current dialogue, pause it for opening a new sub-dialogue, or close the dialogue to possibly start a new dialogue. After this, the agent either abandons the current plan to make a new plan, or adjusts the current plan.

Depending on the type of dialogue, the content of the plan is different. In the case of information-seeking dialogues, the plan contains a set of questions (outcome is answers); in inquiry dialogues the plan contains either a set of defeasible rules or facts (outcome is arguments or new knowledge); in the case of a deliberation dialogue the plan contains a set of potential actions to be decided upon (outcome is actions agreed upon); and in the persuasion dialogues the plan contains a set of statements and arguments related to the dialogue goal, which may be used in a persuasive argumentation dialogue (outcome is a change of priority or belief).

- i When human does not respond as expected : In the case when the human ceases to respond, the Coach Agent puts the dialogue on hold, and defines and stores, if present, partial results of opened dialogues. At a later occasion, the agent can offer the human to resume a dialogue put on hold.
- ii Prioritize topics following the human's prioritization
- iii Rely on older information and reduce number of repeated questions (will be based on learning methods in the future, in order to discover how frequent conditions changes in an individual as well as groups of individuals, e.g. pain conditions)

4.2.3 Emphatic behavior

The Coach Agent uses its ontology (from 4) to associate each action with a concept that is common for both the human and the agents such as *initiating*, *sustaining* and *ending a conversation*. These concepts are used to define certain behaviors that are typically appropriate in a phase of a dialogue, for example, “Hi” is associated to concept “*initiating a conversation*”. The different types of dialogues and dialogue moves described in previous section are also integrated in the agent's model of dialogue activity embedded in the ontology, for instance, the move *affirm*, means that the agent emphasizes and affirms the emotional state, which the human agent expresses, by using different empathic statements such as “*does not sound good*” and “*I see*”. This enables the Coach Agent to simulate the behavior of a participant who listens with some empathy in our user study. The agent distinguishes between positive and negative emotional states in the human by analyzing if the human's response to a question that is associated with one of the two categories “normal/healthy” or “pathologic/not normal” in the ontology. For instance, the answer “yes” may be positive or negative, depending on the context. However, not all values are suitable to be associated in this way, and consequently, the agent's ability to respond in an emphatic way in current implementation is limited.

The Coach Agent acquires new knowledge partly by posing explicit questions in the asymmetric dialogues, where the human is merely responding to questions.

One example is in the situation when the Coach Agent presents a claim about a topic, followed by the move *believe*, which contains the question “*What do you think?*” (ADD EXAMPLE). This is a generic question, which gives the agent some indication if the human agent agrees to the claim or if the agent needs to adjust its strategies for the next move in the dialogue.

4.2.4 Reasoning Strategies

A balance is strived for between the qualities exhaustiveness and efficiency. In any assessments done in healthcare, the optimal would be a complete set of information in order to optimize decisions. However, some information may not be available, time may be limited and the cooperation of the human actor may not persist if the assessment takes too long time, possibly due to limited motivation or capacity. Therefore, strategies have been built into the Coach Agent to do the following for selecting the next move and reducing the number of moves:

- i Prioritize topics following the domain expert's prioritization as structured in the domain knowledge repository
- ii Prioritize topics for which the agent has advice or suggestion of intervention, already in information-seeking dialogues
- iii Identify and rank questions associated to premises based on the frequency they occur in rules
- iv Identify partially satisfied rules, and pose questions to satisfy these before focussing on areas where there are less knowledge
- v Identify questions which are of “screening” type, aimed to find indications of problems. This by identifying rules with conclusions associated to concepts at higher levels of specificity, e.g., “memory” rather than “episodic memory”

The following is the algorithm for the strategies implemented in the dialogue system.

The algorithm depicted in Figure 2, begins with the assumption that there exists at least one rule in the knowledge base for the agent to fetch during the dialogue. This rule has a predicate which has the dialogue topic as the conclusion (refer equation1).

$$\exists r \in RQ[(\sum_{j=1}^m CQ_j, l) \rightarrow P] \quad (1)$$

where, r is a rule, which belongs to R ,
 R is a set of rules in the agents Belief base,
 $Q[(a, b) \rightarrow b]$ is a predicate.

4.3 Flowchart II: "In-between dialogues"

4.3.1 Local Look Ahead

Given the current formula F and the current cause bin CB , then the decision heuristic of the coach agent selects a single question CQ and the next cause bin for its next dialogue move. It tests the immediate implications of asking and not asking a question to the human. For each question, there are 'a' number of answers.

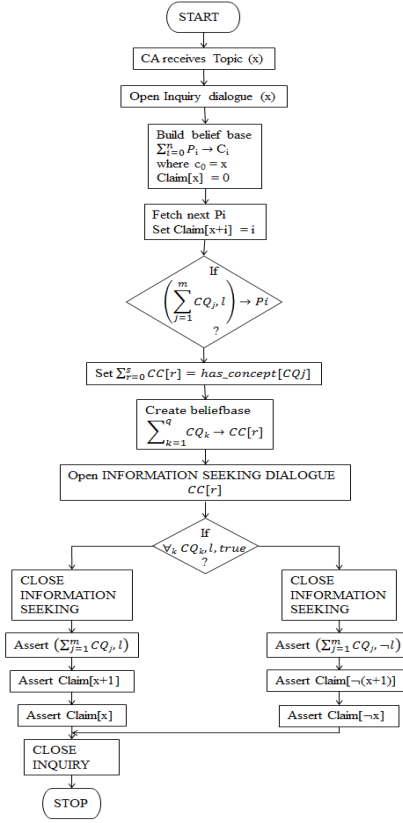


Fig. 2 Flowchart I: "Dialogues": This figure depicts an adaptive dialogue tree, that allows the agent to lead the human through a collaborative decision-making.

Algorithm 1 In-between dialogues algorithm

- 1: **procedure** PRIORITIZE-CONFIRMATION/ELIMINATION OF CAUSE
 - 2: Set $\sum_{r=1}^s CC[r] = has_concept[CQ_j]$ (from Figure 2)
 - 3: where $\sum_{r=1}^s CC[r] \ni [CB_1, CB_2, \dots, CB_n]$ i.e., each Cause Bin (CB) belongs to a concept (CC)
 - 4: create beliefbase

$$F = \sum_{k=1}^q CQ_k \rightarrow CC[r] \quad (2)$$
 - 5: Select next CQ_k such that it confirms/eliminates a CB_n based on lookahead (described in the section 4.3.1)
 - 6: **end procedure**
-

Each answer has an array with 'c' cause weight values for each cause. When an answer is selected, the coach agent takes the corresponding conclusion weight array and adds it to the cause bin.

Local look ahead method, selects the next question to ask, such that it leads to confirmation of most likely cause, in the cause bin. After each dialogue move,

it evaluates the cause bin to select the most likely cause. It then searches for the question which has an answer which leads to confirmation of the cause. If the human selects some other answer than the answer which leads to confirmation of the cause, due to the nature of cause weights assigned to each answer, the evaluated cause bin value will change such a way that current most likely cause will become less likely and some other cause will become most likely. The local look ahead method, repeats the above process to select the next question. This process continues, until a cause is confirmed or all the questions, from the belief base for the selected topic, have been asked. Cause confirmation happens when the most likely cause bin value is above the threshold value w.r.t other cause bin values.

After confirming a cause, the coach agent proposes the same to the human. The human agent can then either agree or disagree to a cause. If the human agrees then the coach agent initiates a deliberation dialogue to come to a solution or action to take for handling the cause. Whereas in the latter case, when the human disagrees to the cause concluded by the agent, the Coach agent continues to maximize the next optimal cause of the dialogue topic. At any point in the dialogue, the human can pause or quit the dialogue.

5 Evaluation Study Results

Each participant was first engaged in a baseline assessment, using the iHelp application in dialogue with a therapist. By answering questions in a structured format, information is collected and a baseline user model including prioritized goals can be formed by the agent based on the information.

Next the participant was instructed to freely explore the dialogue system, where (s)he could select topics. When a topic was selected, the Coach Agent was activated and directed the dialogue.

6 Conclusions

At this point, we adopt strategies based on therapeutic values, however, we envision that these strategies can be further improved and tailored through supplementing with learning methods in the future.

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